Benchmarking continuous optimization techniques for DAG learning

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Domain

- The structure of a Bayesian network can be inferred from data
- However, constructing a DAG has a combinatorial cost
- This makes it very expensive for networks with many variables
- Numerical methods could be applied to get a “good enough” solution in reasonable time

Classical algorithms

- PC[1] - constraint-based: starts from a fully connected graph and removes edges using conditional independence tests
- fGES[2] - score-based: greedily adds edges to maximize a score, then greedily removes them in a backward phase
- The space of possible graphs grows super-exponentially with the number of variables
- Main culprit: acyclicity, without it you could fix parents for each node independently, giving a polynomial bound

[1] Spirtes, Glymour, *An Algorithm for Fast Recovery of Sparse Causal Graphs* [4]

[2] Ramsey, Glymour, Sanchez-Romero, Glymour, *A Million Variables and More: The Fast Greedy Equivalence Search Algorithm for Learning High-Dimensional Graphical Structure of Tetrad* [1]

NOTEARS[3]

- Model the graph as an adjacency matrix W
- Find a **smooth** function h s.t. $h(W) = 0 \iff W$ is an acyclic graph
- Use the Lagrangian method for constrained optimization on W under the constraint $h(W) = 0$

Theorem

$$h(W) := \mathrm{tr}\left(e^{W \circ W}\right) - d = 0 \iff W \text{ is a DAG}$$

where d is the number of variables.

[3] Zheng, Aragam, Ravikumar, Xing, *DAGs with NO TEARS: Continuous Optimization for Structure Learning* [6]

DAG-GNN[4]

- Builds on NOTEARS
- Replaces linear SEM with a VAE using a GNN encoder/decoder
- Captures non-linear dependencies
- Polynomial constraint for acyclicity

Theorem

For any $\alpha > 0$

$$h(W) := \mathrm{tr}\left((\mathcal{I} + \alpha W \circ W)^d\right) - d = 0 \iff W \text{ is a DAG}$$

where d *is the number of variables.*

[4] Yu, Chen, Gao, Yu, *DAG-GNN: DAG Structure Learning with Graph Neural Networks* [5]

Aim

- Compare classical algorithms against NOTEARS and DAG-GNN
- Evaluate how good is the graph reconstructed by each algortihm
- Evaluate resource cost and scaling w.r.t. number of variables and dataset size

Method[5][6]

- Picked 3 networks from the bnlearn repository to generate synthetic dataset + 1 real world dataset
- CANCER, 5 nodes discrete variables
- CHILD, 20 nodes discrete variables
- ALARM, 37 nodes discrete variables
- Sachs et al., 11 nodes continuous variables

[5] Scutari, *Bayesian Network Repository* [3]

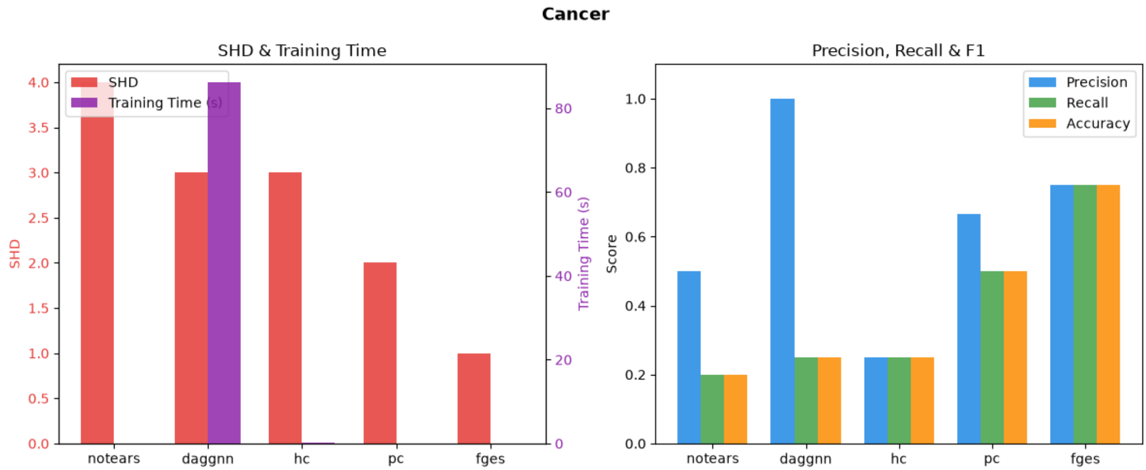
[6] Sachs, Perez, Pe'er, Lauffenburger, Nolan, *Causal Protein-Signaling Networks Derived from Multiparameter Single-Cell Data* [2]

Results

- NOTEARS performs really well adding a tolerable overhead
- DAG-GNN failed to beat NOTEARS and takes significantly longer
- They can scale better to more variables (eg compared to fGES)
- They scale worse with a bigger dataset (other algorithms are almost unaffected)

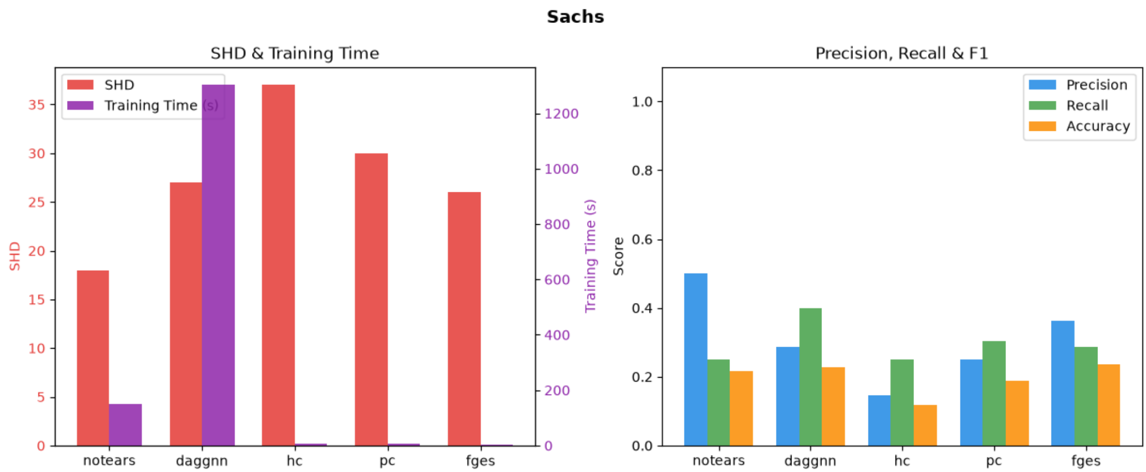
Algorithm	Time (s)	nSHD	Cancer	Sachs	Child	Alarm
NOTEARS	57.27	0.90	4	18	21	34
DAG-GNN	450.88	1.00	3	27	24	42
HC	10.37	1.30	3	37	30	61
PC	2.99	0.90	2	30	26	29
FGES	30.87	1.00	1	26	27	54

Cancer



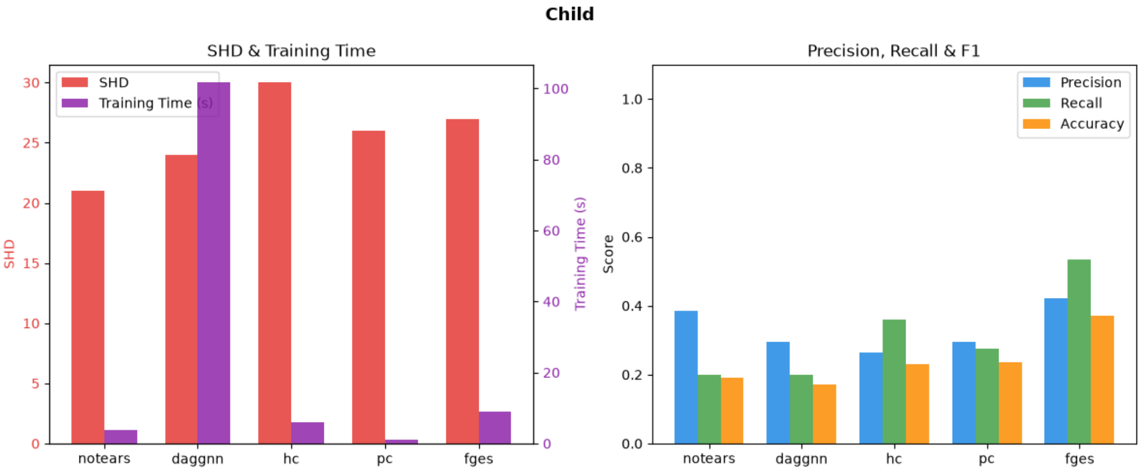
Cancer
SHD & Training Time
SHD
Training Time (s)
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
80
60
40
20
0
notears
daggnn
hc
pc
fges
Precision, Recall & F1
Precision
Recall
Accuracy
Score
1.0
0.8
0.6
0.4
0.2
0.0
notears
daggnn
hc
pc
fges

Sachs



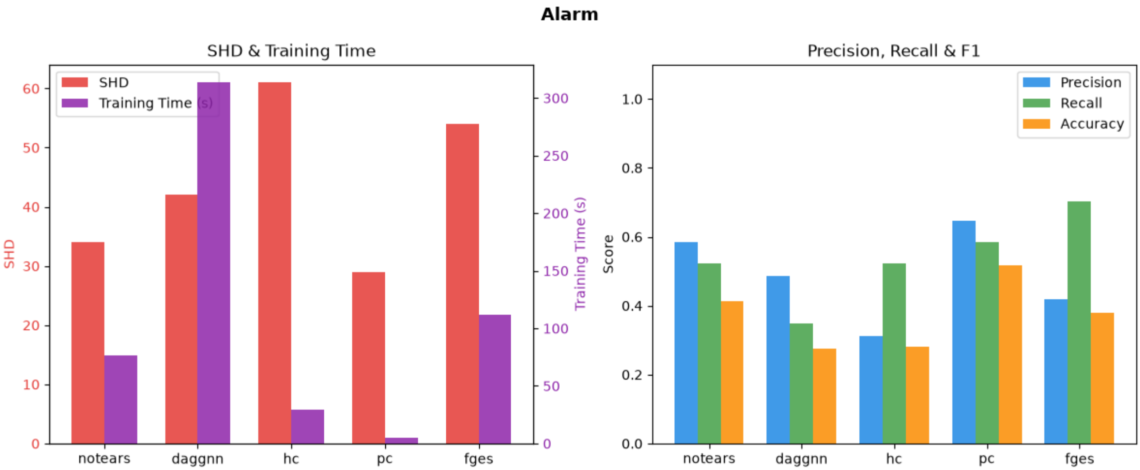
Sachs
SHD & Training Time
SHD
Training Time (s)
SHD
Training Time (s)
0
5
10
15
20
25
30
35
0
200
400
600
800
1000
1200
notears
daggnn
hc
pc
fges
Precision, Recall & F1
Precision
Recall
Accuracy
Score
0.0
0.2
0.4
0.6
0.8
1.0
notears
daggnn
hc
pc
fges

Child



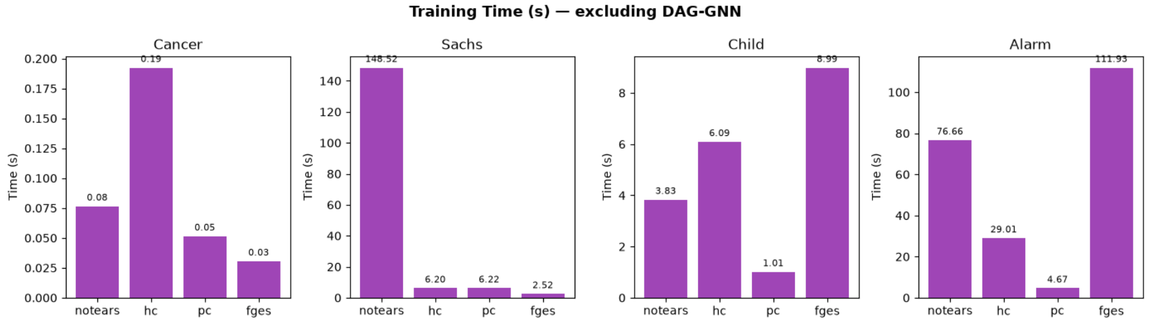
Child
SHD & Training Time
SHD
Training Time (s)
30
25
20
15
10
5
0
100
80
60
40
20
0
notears
daggnn
hc
pc
fges
Precision, Recall & F1
Precision
Recall
Accuracy
Score
1.0
0.8
0.6
0.4
0.2
0.0
notears
daggnn
hc
pc
fges

Alarm



Alarm
SHD & Training Time
SHD
Training Time (s)
60
50
40
30
20
10
0
300
250
200
150
100
50
0
notears
daggnn
hc
pc
fges
Precision, Recall & F1
Precision
Recall
Accuracy
Score
1.0
0.8
0.6
0.4
0.2
0.0
notears
daggnn
hc
pc
fges

Execution times minus DAG-GNN



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